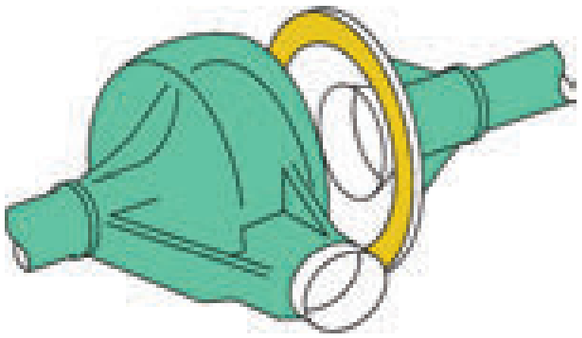


Figure 4.11.2 Split type differential housing

DO YOU KNOW?

In 1892, Rudolf Diesel a German engineer invented a "New Rational Combustion Engine" which lead him to building the first Diesel Engine in 1897.



4.12 Rear axle

The engine's power comes from the differential is transmitted through the rigid shaft called as the rear axle. It is made as two parts such as left side and right side in the vehicle and used. The axle's inner edge is connected with the sun gear by the splines and outer edge is connected to the wheel hub. This axle withstands the weight, vibrations and torque of the vehicles.

Purpose

1. To withstand rear side weight of the vehicle.
2. To transmit the engine power to rear wheels.
3. To act as axle needle for the wheels.
4. It acts as housing for the final drive, differential and half shafts.

Forces acting on the rear axle

The following forces act on the rear axle

1. Torque reaction
2. Driving torque or driving thrust
3. Side thrust
4. Weight of the body

1. Torque reaction

When the torque is transmitted from the propeller shaft through the differential, the bevel pinion is started to rotate over the crown wheel, at the time an opposite force is developed against the direction of differential housing. This force is equal to driving torque and operates in the opposite direction. This force produces a bending moment on the propeller shaft.

2. Driving torque

It is the torque transferred from the engine to the axle casing. The force created due to this on the rear axle is transferred to the body through chassis frame. Radius rods in the chassis are used to transmit this drive torque to the body.

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